

09d — The N+1 Problem

Key Concepts

Term	Definition
N+1 Problem	1 query fetches a list, then N more queries fire for each item's related data
Lazy Loading	Related data fetched on demand — triggers a query each time you access a relationship
Eager Loading	Related data fetched upfront in the same query — eliminates N+1
DataLoader	Batching pattern (used in GraphQL) — groups N queries into 1

How It Happens

```
Query 1:  SELECT * FROM orders           → returns 100 orders
Query 2:  SELECT * FROM users WHERE id = 1
Query 3:  SELECT * FROM users WHERE id = 2
...
Query 101: SELECT * FROM users WHERE id = 100

Total: 101 queries instead of 1
```

Why ORMs Cause It

- ORMs default to **lazy loading**
- Accessing a relationship inside a loop triggers a new query per iteration

```
// BAD — N+1
var orders = context.Orders.ToList();
foreach (var order in orders) {
    Console.WriteLine(order.User.Name); // new query each iteration
}

// GOOD — eager loading
var orders = context.Orders.Include(o => o.User).ToList(); // 1 query
```

How to Fix

Fix	How
Eager loading	<code>.Include()</code> in EF Core

Fix	How
Raw SQL JOIN	Write the join yourself
DataLoader	Batch queries — used in GraphQL APIs
Select only needed fields	Avoid loading full objects unnecessarily

How to Detect

- Query count grows proportionally with list size
- EF Core query logging — see repeated similar queries
- MiniProfiler — shows query count per request
- SQL Profiler / Datadog APM — spots repeated patterns in production

Industry Examples

- **Shopify** — N+1 is their most common Rails performance issue; use **bullet** gem to detect it
- **GitHub** — uses DataLoader in GraphQL API to batch relationship queries
- **Any list page** — orders + customer names, posts + author info = classic N+1 trap

Interview Questions

Question	One-Line Answer
What is the N+1 problem?	1 query for a list + N queries for each item's related data — instead of 1 join
Why do ORMs cause it?	Lazy loading fetches related data on demand — one query per loop iteration
How do you fix it?	Eager loading (.Include()), explicit JOIN, or DataLoader batching
How do you detect it?	Query logging, MiniProfiler, APM tools — look for repeated similar queries
Lazy vs eager loading?	Lazy = on demand (causes N+1); Eager = upfront in same query (prevents N+1)